

Let's assume I am sending a file from **my laptop** to **my friend's computer**, which is located in a different subnet.

Device	IP Address
My Laptop	192.168.1.10/24
Router 1 (Default Gateway)	192.168.1.1
Router 1 to Router 2	10.0.0.1
Router 2 to Router 1	10.0.0.2
Friend's Computer	192.168.2.20/24
Router 2 (Friend's Gateway)	192.168.2.1

Step 1: Packet is Created

I want to send data from my laptop (**192.168.1.10**) to my friend's computer (**192.168.2.20**).

My laptop first checks whether the destination IP belongs to the same subnet.

- My subnet: **192.168.1.0/24**
- Destination: **192.168.2.20**

Since the destination is in a different subnet, the laptop sends the packet to its default gateway (**192.168.1.1**).

Step 2: Router 1 Makes a Routing Decision

Router 1 receives the packet and checks its routing table.

Router 1 Routing Table

Destination Network	Next Hop	Interface
192.168.1.0/24	Directly Connected	LAN
192.168.2.0/24	10.0.0.2	Serial0/0
0.0.0.0/0	ISP	WAN

Router 1 sees that **192.168.2.20** belongs to the **192.168.2.0/24** network.

So, it forwards the packet to the next-hop router at **10.0.0.2**.

Step 3: Router 2 Makes Another Routing Decision

Router 2 receives the packet and checks its routing table.

Router 2 Routing Table

Destination Network	Next Hop	Interface
192.168.2.0/24	Directly Connected	LAN
192.168.1.0/24	10.0.0.1	Serial0/0

Router 2 finds that the destination network (**192.168.2.0/24**) is directly connected.

It forwards the packet through its LAN interface to my friend's computer (**192.168.2.20**).

Step 4: Packet Reaches the Destination

My friend's computer receives the packet successfully.

If my friend replies, the return packet follows the reverse path:

Friend's Computer → Router 2 → Router 1 → My Laptop

Packet Flow

My Laptop

192.168.1.10

|



Router 1

192.168.1.1

|

| 10.0.0.1 ————— 10.0.0.2



Router 2

192.168.2.1

|



Friend's Computer

192.168.2.20